

The Laplace approximation of $\log p(r)$

1 The quantity

At a query image the GP gives a Gaussian posterior on the log firing rate g :

$$g \sim \mathcal{N}(\mu_g, \sigma_g^2), \quad \mu_g = A \bar{\lambda} + \lambda_0, \quad \sigma_g^2 = A^2 \text{var}(\lambda).$$

Given g the spike count is Poisson with rate e^g . The predictive probability of r spikes marginalises over g :

$$p(r) = \int_{-\infty}^{\infty} \underbrace{\frac{e^{rg} e^{-e^g}}{r!}}_{\text{Poisson}(r; e^g)} \underbrace{\frac{1}{\sqrt{2\pi\sigma_g^2}} e^{-(g-\mu_g)^2/2\sigma_g^2}}_{\mathcal{N}(g; \mu_g, \sigma_g^2)} dg.$$

This integral has no closed form.

2 The integrand and its peak

Write the integrand as $e^{\psi(g)}$ up to constants, with

$$\psi(g) = r g - e^g - \frac{(g - \mu_g)^2}{2\sigma_g^2}.$$

ψ is strictly concave, so the integrand has a single peak at $\bar{g}$, where $\psi'(\bar{g}) = 0$:

$$r = e^{\bar{g}} + \frac{\bar{g} - \mu_g}{\sigma_g^2}.$$

This solves in closed form through the principal Lambert- W function:

$$\bar{g}(r) = r\sigma_g^2 + \mu_g - W_0(\sigma_g^2 e^{r\sigma_g^2 + \mu_g}).$$

(The code evaluates W_0 in log-space, so the exponential is never formed.)

3 The approximation

Laplace replaces ψ by its second-order Taylor expansion at the peak,

$$\psi(g) \approx \psi(\bar{g}) + \frac{1}{2} \psi''(\bar{g}) (g - \bar{g})^2,$$

i.e. a Gaussian, whose integral is analytic. Carrying the constants through:

$$\log p(r) \approx r \bar{g} - e^{\bar{g}} - \frac{(\bar{g} - \mu_g)^2}{2\sigma_g^2} - \frac{1}{2} \log(1 + \sigma_g^2 e^{\bar{g}}) - \log r!$$

Every term is evaluated *exactly* at $\bar{g}$. This expression is the output the algorithm uses, and the object whose accuracy we test.

4 Where the error comes from

The error is exactly what the quadratic expansion drops: the third and higher derivatives of ψ at the peak (its asymmetry). Split ψ into its two factors and differentiate:

$$\psi(g) = \underbrace{rg - e^g}_{\text{Poisson}} \underbrace{- (g - \mu_g)^2 / 2\sigma_g^2}_{\text{Normal}}, \quad \begin{aligned} \psi''(g) &= -e^g - 1/\sigma_g^2, \\ \psi'''(g) &= -e^g, \\ \psi''''(g) &= -e^g. \end{aligned}$$

The Normal factor is exactly quadratic: it adds the constant $-1/\sigma_g^2$ to the curvature and *zero* to every higher derivative. The Gaussian fit reproduces it exactly, so it contributes nothing to the error. All of the error comes from the $-e^g$ (Poisson) term, the only non-quadratic part of ψ .

5 Why σ_g^2 controls the error

σ_g^2 enters only through the curvature at the peak,

$$|\psi''(\bar{g})| = e^{\bar{g}} + \frac{1}{\sigma_g^2},$$

which sets the width of the fitted Gaussian, $1/\sqrt{|\psi''(\bar{g})|}$.

Because the integrand's leading factor is the Gaussian $e^{\frac{1}{2}\psi''(\bar{g})(g-\bar{g})^2}$, its weight lies within this width of the peak. The dropped cubic remainder $\frac{1}{6}\psi'''(\bar{g})(g-\bar{g})^3 + \dots$ — a fixed coefficient $\psi'''(\bar{g}) = -e^{\bar{g}}$ times $(g-\bar{g})^3$ — therefore matters only at $(g-\bar{g})$ of that size, where its magnitude is

$$\frac{1}{6} \frac{|\psi'''(\bar{g})|}{|\psi''(\bar{g})|^{3/2}} = \frac{1}{6} \frac{e^{\bar{g}}}{(e^{\bar{g}} + 1/\sigma_g^2)^{3/2}}.$$

So the asymmetry is measured against the curvature, and σ_g^2 enters the error only through this width, leaving ψ''' untouched.

- Small σ_g^2 : $1/\sigma_g^2$ dominates, the peak is narrow, and over that width e^g barely curves — its asymmetry is invisible and the approximation is near-exact.
- Large σ_g^2 : the confinement weakens, the peak widens, and the asymmetry of e^g is exposed — the error grows.

So σ_g^2 is the knob and the Poisson e^g is the source: the prior does not cause the error, it suppresses it by pinning the peak.

The difference $\log p_{\text{true}}(r) - \log p_{\text{Laplace}}(r)$ is what `laplace_pointwise_error.py` measures against an adaptive-quadrature reference.